

false

Logical 0 (false)

Syntax

```
false
F = false(n)
F = false(sz)
F = false(sz1,...,szN)
F = false(__,like=p)
```

Description

false is shorthand for the logical value 0. [example](#)

F = false(**n**) is an n-by-n array of logical zeros. [example](#)

F = false(**sz**) is an array of logical zeros where the size vector, **sz**, defines size(**F**). For example, false([2 3]) returns a 2-by-3 array of logical zeros. [example](#)

F = false(**sz1**,...,**szN**) is a **sz1**-by-...-by-**szN** array of logical zeros where **sz1**,...,**szN** indicates the size of each dimension. For example, false(2,3) returns a 2-by-3 array of logical zeros. [example](#)

F = false(__,like=**p**) returns an array of logical zeros of the same sparsity as the logical variable **p** using any of the previous size syntaxes. [example](#)

Examples

[collapse all](#)

Generate Square Matrix of Logical Zeros

Use false to generate a 3-by-3 square matrix of logical zeros.

[Open in MATLAB Online](#)[Copy Command](#)

```
A = false(3)
```

[Get](#) ▼

A = 3×3 logical array

```
0  0  0
0  0  0
0  0  0
```

```
class(A)
```

[Get](#) ▼

```
ans =  
'logical'  
The result is of class logical.
```

Generate Array of Logical Zeros with Arbitrary Dimensions

Use false to generate a 3-by-2-by-2 array of logical zeros.

[Open in MATLAB Online](#)[Copy Command](#)

```
false(3,2,2)
```

[Get](#) ▼

```
ans = 3x2x2 logical array  
ans(:,:,1) =
```

```
0 0  
0 0  
0 0
```

```
ans(:,:,2) =
```

```
0 0  
0 0  
0 0
```

Alternatively, use a size vector to specify the size of the matrix.

```
false([3 2 2])
```

[Get](#) ▼

```
ans = 3x2x2 logical array  
ans(:,:,1) =
```

```
0 0  
0 0  
0 0
```

```
ans(:,:,2) =
```

```
0 0  
0 0  
0 0
```

Note that specifying multiple vector inputs returns an error.

Execute Logic Statement

false along with true can be used to execute logic statements.

Test the logical statement

$\sim(A \text{ and } B) = (\sim A) \text{ or } (\sim B)$

for $A = \text{false}$ and $B = \text{true}$.

[Open in MATLAB Online](#)
[Copy Command](#)

```
~(false & true) == (~false) | (~true)
```

[Get](#)

```
ans = logical
      1
```

The result is logical 1 (true), since the logical statements on both sides of the equation are equivalent. This logical statement is an instance of De Morgan's Law.

Generate Logical Array of Selected Sparsity

Generate a logical array of the same data type and sparsity as the selected array.

[Open in MATLAB Online](#)
[Copy Command](#)

```
A = logical(sparse(5,3));
whos A
```

[Get](#)

Name	Size	Bytes	Class	Attributes
A	5x3	41	logical	sparse

```
F = false(4,like=A);
whos F
```

[Get](#)

Name	Size	Bytes	Class	Attributes
F	4x4	49	logical	sparse

The output array F has the same sparse attribute as the specified array A.

Input Arguments

[collapse all](#)

n — Size of square matrix
integer

Size of square matrix, specified as an integer. n sets the output array size to n-by-n. For example, false(3) returns a 3-by-3 array of logical zeros.

- If n is 0, then F is an empty matrix.
- If n is negative, then it is treated as 0.

Data Types: int8 | int16 | int32 | int64 | uint8 | uint16 | uint32 | uint64

sz — Size vector
row vector of integers

Size vector, specified as a row vector of integers. For example, `false([2 3])` returns a 2-by-3 array of logical zeros.

- If the size of any dimension is 0, then `F` is an empty array.
- If the size of any dimension is negative, then it is treated as 0.
- If any trailing dimensions greater than 2 have a size of 1, then the output, `F`, does not include those dimensions.

Data Types: int8 | int16 | int32 | int64 | uint8 | uint16 | uint32 | uint64

sz1, ..., szN — Size inputs
comma-separated list of integers

Size inputs, specified by a comma-separated list of integers. For example, `false(2,3)` returns a 2-by-3 array of logical zeros.

- If the size of any dimension is 0, then `F` is an empty array.
- If the size of any dimension is negative, then it is treated as 0.
- If any trailing dimensions greater than 2 have a size of 1, then the output, `F`, does not include those dimensions.

Data Types: int8 | int16 | int32 | int64 | uint8 | uint16 | uint32 | uint64

p — Prototype
logical variable

Prototype, specified as a logical variable.

Data Types: single | double | int8 | int16 | int32 | int64 | uint8 | uint16 | uint32 | uint64

Output Arguments

[collapse all](#)

F — Output of logical zeros
scalar | vector | matrix | N-D array

Output of logical zeros, returned as a scalar, vector, matrix, or N-D array.

Data Types: logical

Tips

- `false(n)` is much faster and more memory efficient than `logical(zeros(n))`.

Extended Capabilities

[expand all](#)

C/C++ Code Generation

Generate C and C++ code using MATLAB® Coder™.

GPU Code Generation

Generate CUDA® code for NVIDIA® GPUs using GPU Coder™.

Thread-Based Environment

Run code in the background using MATLAB® backgroundPool or accelerate code with Parallel Computing Toolbox™ ThreadPool.

GPU Arrays

Accelerate code by running on a graphics processing unit (GPU) using Parallel Computing Toolbox™.

Distributed Arrays

Partition large arrays across the combined memory of your cluster using Parallel Computing Toolbox™.

Version History

Introduced before R2006a

See Also

[true](#) | [logical](#)

Topics

[Class Support for Array-Creation Functions](#)

YOU MIGHT ALSO BE INTERESTED IN

[Converting Deep Learning Models Between PyTorch, TensorFlow, and MATLAB](#)

Import and export AI models between PyTorch®, TensorFlow™, and MATLAB®.

[Learn more](#)